Gray Code

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2205 | Introductory Problems | https://cses.fi/problemset/task/2205

A Gray code is a list of all 2^n bit strings of length n, where any two successive strings differ in exactly one bit (i.e., their Hamming distance is one).

Your task is to create a Gray code for a given length n.

Input

The only input line has an integer n.

Output

Print 2^n lines that describe the Gray code. You can print any valid solution.

Constraints

- $1 \le n \le 16$

Example

Input
2

Output
00 01 11 10